PassFrame
tuition

# Number Codes

Year 5 Accelerated — 11+ Lesson Worksheet (PassFrame Tuition)

**Worked examples — tutor talks through these first**

**A) Easy:** If the code for 12 is 21 (the digits are reversed), what is the code for 34?
*Method & answer: Reverse the digits of 34: 43. Answer: 43.*

**B) Standard:** If each digit shifts forward by 1 (wrapping 9 back round to 0), 123 becomes 234. What is the code for 567?
*Method & answer: Shift every digit of 567 forward by 1: 5→6, 6→7, 7→8. Answer: 678.*

**C) Advanced:** If 3 # 4 = 10 (the rule is: double the first number, then add the second), what is 5 # 2?
*Method & answer: Apply the same rule to 5 and 2: (2×5) + 2 = 12. Answer: 12.*

## Easy / Standard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. If the code for 13 is 31 (the digits are reversed), what is the code for 46?
2. If the code for 56 is 65 (the digits are reversed), what is the code for 78?
3. If each digit shifts forward by 1 (wrapping 9 back round to 0), 11 becomes 22. What is the code for 55?
4. If each digit shifts forward by 2 (wrapping 9 back round to 0), 23 becomes 45. What is the code for 67?
5. If each digit shifts back by 1 (wrapping 0 back round to 9), 35 becomes 24. What is the code for 67?
6. If the code for 24 is 42 (the digits are reversed), what is the code for 68?
7. If each digit shifts forward by 1 (wrapping 9 back round to 0), 304 becomes 415. What is the code for 213?
8. If the code for 23 is 5 (the digits are added together), what is the code for 45?
9. If the code for 31 is 4 (the digits are added together), what is the code for 52?
10. If each digit shifts forward by 2 (wrapping 9 back round to 0), 456 becomes 678. What is the code for 123?
11. If the code for 19 is 91 (the digits are reversed), what is the code for 82?
12. If each digit shifts back by 2 (wrapping 0 back round to 9), 59 becomes 37. What is the code for 84?
13. If the code for 14 is 5 (the digits are added together), what is the code for 26?
14. If 8 # 3 = 11 (the rule is: add the two numbers), what is 6 # 5?
15. If 9 # 2 = 7 (the rule is: subtract the second number from the first), what is 10 # 4?

## Standard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. If each digit shifts forward by 4 (wrapping 9 back round to 0), 2345 becomes 6789. What is the code for 6789?
2. If the code for 1234 is 4321 (the
6. If 11 # 4 = 36 (the rule is: multiply the two numbers, then subtract the second number's double), what is 13 # 5?
7. If the code for 4072 is 2704 (the
11. If each digit shifts back by 5 (wrapping 0 back round to 9), 604 becomes 159. What is the code for 839?
12. If 8 # 9 = 34 (the rule is: add the

digits are reversed), what is the code for 5678?

**3.** If the code for 482 is 14 (the digits are added together), what is the code for 591?

**4.** If each digit shifts back by 3 (wrapping 0 back round to 9), 6048 becomes 3715. What is the code for 8271?

**5.** If 12 # 5 = 29 (the rule is: double the first number, then add the second), what is 18 # 7?

digits are reversed), what is the code for 6395?

**8.** If each digit shifts forward by 5 (wrapping 9 back round to 0), 148 becomes 693. What is the code for 637?

**9.** If the code for GARDEN is 49 (add up the alphabet positions: G=7, A=1, R=18, D=4, E=5, N=14, so 7+1+18+4+5+14=49), what is the code for FOREST using the same rule?

**10.** If the code for 674 is 17 (the digits are added together), what is the code for 829?

two numbers, then double the result), what is 12 # 15?

**13.** If the code for 3805 is 5083 (the digits are reversed), what is the code for 2081?

**14.** If each digit shifts forward by 4 (wrapping 9 back round to 0), 7182 becomes 1526. What is the code for 3456?

**15.** If the code for BRIDGE is 45 (add up the alphabet positions: B=2, R=18, I=9, D=4, G=7, E=5, so 2+18+9+4+7+5=45), what is the code for TUNNEL using the same rule?

## Standard / Hard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

**1.** A code shifts the 1st digit forward by 1, the 2nd digit forward by 2, and the 3rd digit forward by 3 (wrapping round if needed). Using this rule, 123 becomes 246 (1+1=2, 2+2=4, 3+3=6). Work out what the same rule turns 214 into.

**2.** A code reverses the digits of a number, then adds 1 to the result. Applying this to 23: reverse to get 32, then add 1 to get 33. What does the same rule turn 45 into?

**3.** If 4 # 3 = 16 (the rule is: multiply the two numbers, then add the first number back on), what is 5 # 2?

**4.** A code adds the two digits of a number together, then doubles the result. Applying this to 23: 2+3=5, then 5×2=10. What does the same

**6.** If the code for CAT is 60 (multiply the alphabet positions: C=3, A=1, T=20, so 3×1×20=60), what is the code for DOG?

**7.** A code reverses a number's digits, then subtracts the original number's digit sum. Applying this to 41: reverse to 14, digit sum of 41 is 5, so 14−5=9. What does the same rule turn 62 into?

**8.** A code uses both numbers differently depending on which is bigger: if the second number is bigger, multiply them; if the first is bigger (or they are equal), add them. Using this rule, 4 # 6 = 24 (6 is bigger, so multiply: 4×6=24) and 9 # 3 = 12 (9 is bigger, so add: 9+3=12). What is 5 # 8?

**9.** A code swaps the 1st and 3rd digits of a 3-digit number and leaves the middle digit the same. Applying this rule, 132 becomes 231. What

**11.** A code adds the digit sum of a number to the number itself. Applying this to 23: digit sum is 5, so 23+5=28. What does the same rule turn 47 into?

**12.** If 3 # 2 = 11 and 5 # 4 = 29 (the rule squares the first number and adds the second), what is 4 # 3?

**13.** A code shifts the digits of a number forward by 1, then reverses the whole number. Applying this to 12: shift to get 23, then reverse to get 32. What does the same rule turn 45 into?

**14.** A code multiplies the two digits of a 2-digit number together. Applying this to 26: 2×6=12. What does the same rule turn 39 into?

rule turn 47 into?

**5.** A code shifts every digit forward by 5, wrapping round past 9 back to 0. Using this code, what does 48 become?

does the same rule turn 456 into?

**10.** A code takes a 2-digit number, doubles each digit, and wraps any result of 10 or more back round by subtracting 10. Applying this to 37: 3×2=6, 7×2=14→14−10=4, giving 64. What does the same rule turn 68 into?

**15.** If the code for CAT is 22 (the rule is: add up the alphabet positions, then subtract 2, since 3+1+20−2=22), what is the code for DOG?